IMDb Movie Analysis

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The Problem Statement

The impact of this problem is significant for movie producers, directors, and investors who want to understand what makes a movie successful to make informed decisions in their future projects.

Let us understand the data and the question.

And the Question is...

"What factors influence the success of a movie on IMDB?"

Let us start the analysis to find the solution to this...

Data Preprocessing

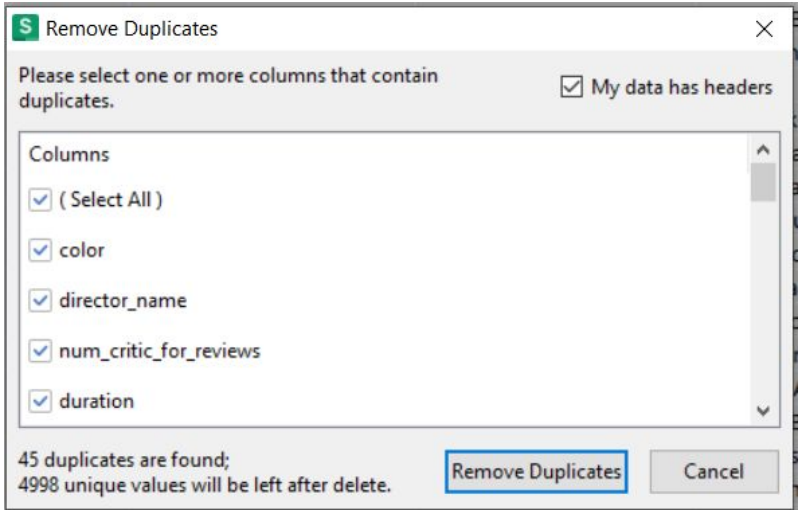
Data Preprocessing

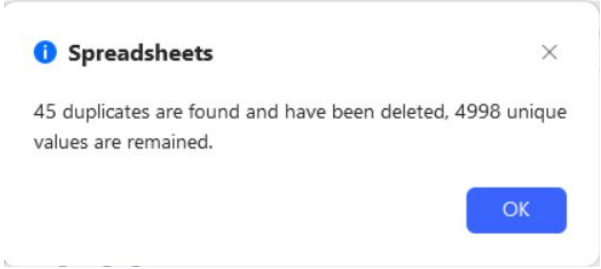
Data Preprocessing includes preparing the data for analysis and deriving insights. Its steps include:

1. **Handling Missing Values in Data**
2. **Removing Duplicate values**
3. **Deciding which columns to keep**
4. **Converting values to correct data type**
5. **Visualizing the data**

1. Removing Duplicates

- I wanted to remove duplicates before proceeding with Missing data
- I selected the data using Ctrl+A, Went to Data Tab and then clicked on Handle Duplicate Data Button ↠ Remove Duplicates
- I found 45 Duplicate values for all rows, I decided to remove them as they will skew the analysis ahead.





2. Decide Columns that are useful in Analysis

I will remove unnecessary columns in data before proceeding with Missing data

1. **color**: This indicates whether the movie is in color or black-and-white. It is unlikely to significantly impact modern movie ratings.
2. **director_facebook_likes**: The number of likes on a director's Facebook page may not directly correlate with the movie's IMDB score.
3. **actor_3_facebook_likes**: Similar to the director's Facebook likes, this may not have a direct impact on the movie's rating.
4. **facenumber_in_poster**: The number of faces in the movie poster is unlikely to influence the movie's IMDB rating.
5. **plot_keywords**: While plot keywords can be used for text analysis, but not relevant here for numerical analysis.
6. **movie_imdb_link**: The URL link to the movie's IMDB page is not useful for the analysis.
7. **aspect_ratio**: The aspect ratio of the movie might have limited impact on the movie's success or IMDB rating.

Removing columns from Data

List of remaining columns in Data using Python

- director_name
- Num_critic_for_reviews
- Duration
- actor_2_name
- actor_1_facebook_likes
- gross
- genres
- actor_1_name
- movie_title
- Num_voted_users
- Cast_total_facebook_likes
- actor_3_name
- num_user_for_reviews
- language
- Country
- content_rating
- budget
- title_year
- actor_2_facebook_likes
- imdb_score
- movie_facebook_likes

3. Handling Missing Data

From the data that remains I will now Filter it and remove the rows containing Blanks, Hyphens, Nan values etc

- For columns such as - director_name, num_critic_for_reviews, duration, gross, genres, actor_1_name, movie_title and language I decided to Remove the blank value rows completely
- For column content_rating I decided to rename the Blank columns as "Unrated"
- For the column budget, I decided to take the median of the budget column as it won't be affected by any outliers, but I did not get representative value, so decided to Delete those rows as well

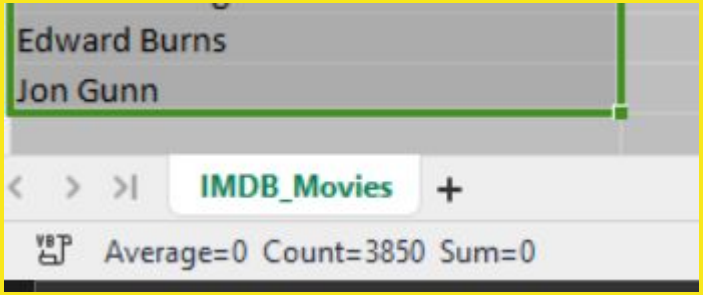
3. Handling Missing Data continued

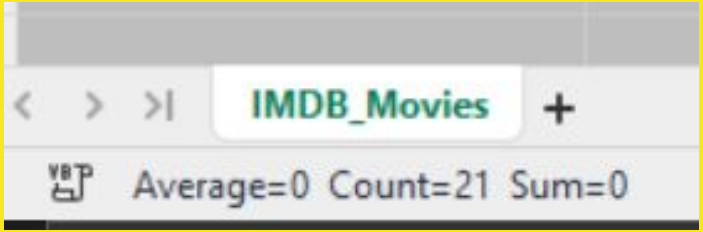
- For columns such as - actor_2_name, actor_1_facebook_likes, actor_2_facebook_likes, movie_facebook_likes - I decided to keep the blanks as it might happen that some movie does not have a second Lead or Actor, also not all actors use facebook

  Now the Missing Data Handling is over, and I will start with further analysis

After removing missing values

Data count becomes
3850 rows and 21 columns





4. Converting Data types

Ensuring that the columns like duration, budget, gross, etc., are in numeric format.

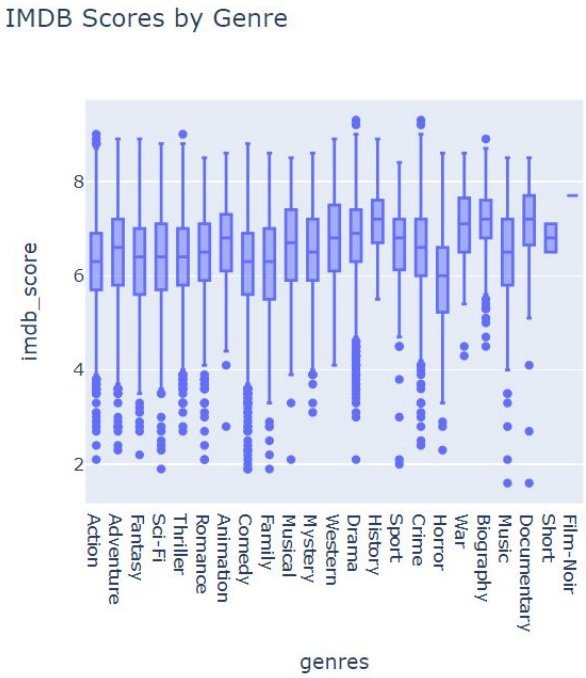
- Select the column, right-click, and choose Format Cells to set the correct data type.
- Converted actor_1_facebook_likes, actor_2_facebook_likes, movie_facebook_likes columns to number with 1000 separator and 0 decimal values and gross and budget columns to currency data type

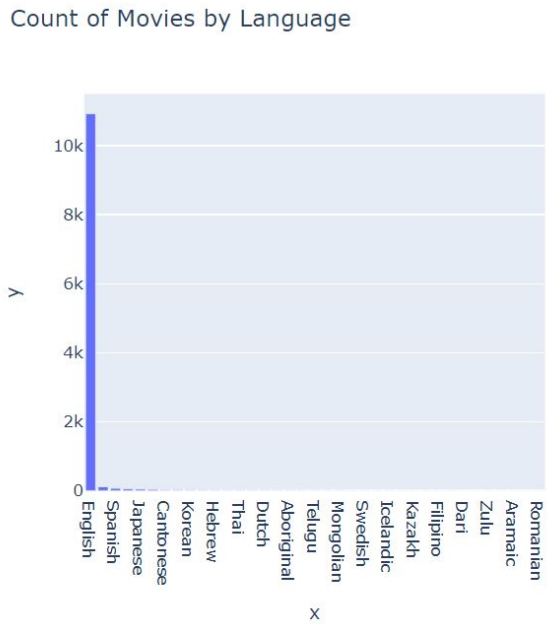
4. Converting Data types continued

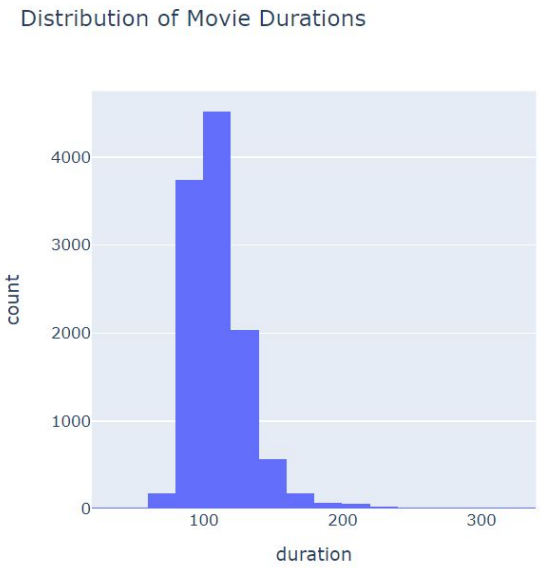
Q	R	S	T	U
budget	title_year	actor_2_facebook_likes	imdb_scor	movie_facebook_likes
$25,00,00,000	2016	4,000	6.9	1,97,000
$20,90,00,000	2006	10,000	6.1	0
$20,00,00,000	2008	412	6.7	0
$22,50,00,000	2006	5,000	7.3	5,000
$21,50,00,000	2013	2,000	6.5	48,000
$22,50,00,000	2013	3,000	7.2	1,18,000
$22,50,00,000	2008	216	6.6	0
$22,00,00,000	2012	21,000	8.1	1,23,000
$25,00,00,000	2011	11,000	6.7	58,000

5. Visualize the Data

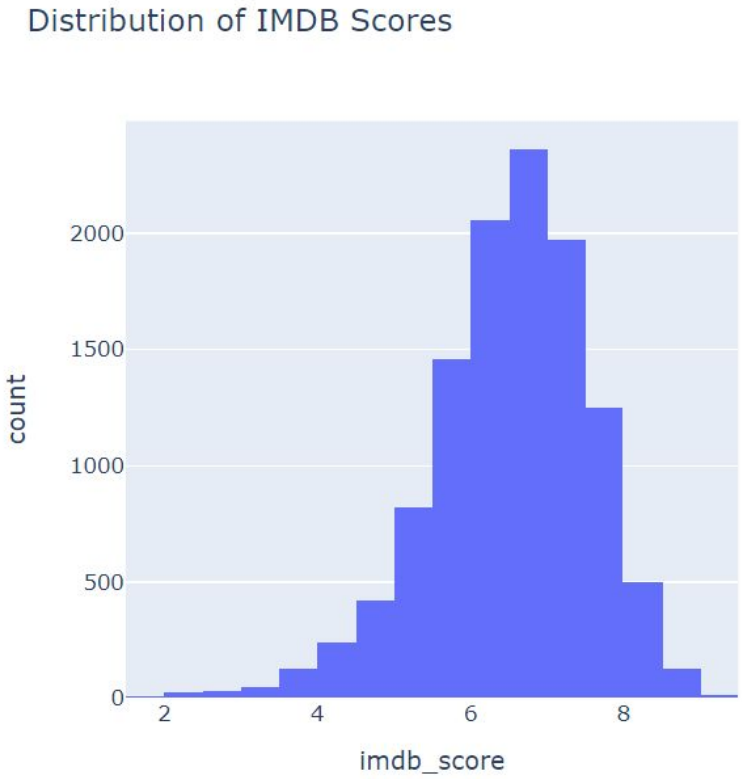
I have used python programming for EDA on the IMDB data:

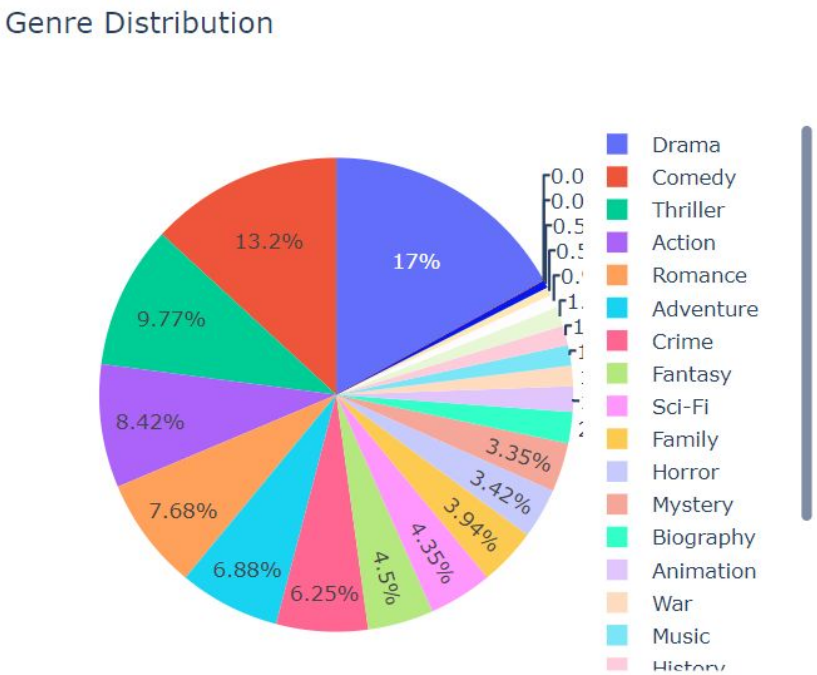






5. Visualize the Data continued





The Five Why's Approach

The Five Why's Approach

Problem Statement **1** :

"Why do movies with higher budgets tend to have higher gross earnings?"

Question	Explanation
1. **Why do movies with higher budgets tend to have higher gross earnings?**	1. Higher budgets allow for better marketing and wider distribution.
2. **Why do better marketing and wider distribution lead to higher gross earnings?**	2. More people become aware of the movie and have access to it, increasing ticket sales.
3. **Why does increased awareness and access boost ticket sales?**	3. Effective marketing creates anticipation and interest, while wide distribution ensures availability.
4. **Why do anticipation and availability drive ticket sales?**	4. People are more likely to watch a movie they are excited about and can easily find in theaters.
5. **Why does excitement and accessibility result in higher earnings?**	5. Positive buzz and ease of access lead to more viewers, higher sales, and thus greater earnings.

Solution:

- **Invest in comprehensive marketing campaigns**: Allocate a significant portion of the budget to marketing to build anticipation.
- **Ensure wide distribution**: Partner with major distribution channels to make the movie accessible to a larger audience.

The Five Why's Approach continued

Problem Statement **2** :

"Why do movies directed by certain directors have higher average IMDB scores?"

Question	Explanation
1. **Why do movies directed by certain directors have higher average IMDB scores?**	1. Some directors consistently produce high-quality films.
2. **Why do these directors consistently produce high-quality films?**	2. They have a unique vision and expertise that resonates with audiences.
3. **Why does their vision and expertise resonate with audiences?**	3. They understand audience preferences and excel at storytelling and visual presentation.
4. **Why do storytelling and visual presentation matter?**	4. Good storytelling and visuals enhance the viewer's experience and satisfaction.
5. **Why does enhanced viewer experience lead to higher scores?**	5. Satisfied viewers are more likely to give positive reviews and high ratings.

Solution:

- **Identify and emulate successful directors' techniques**: Study the methods and styles of successful directors to understand what makes their movies highly rated.
- **Invest in director development**: Provide resources and training to up-and-coming directors to hone their skills and vision.

Data Analytics Tasks

Data Analytics Tasks

1. **Movie Genre Analysis**
2. **Movie Duration Analysis**
3. **Language Analysis: Situation**
4. **Director Analysis: Influence of directors on movie ratings.**
5. **Budget Analysis**

1. Movie Genre Analysis

The objective of the movie genre analysis is to understand the distribution of movie genres in the dataset and analyze their impact on the IMDB score. This involves calculating descriptive statistics (mean, median, mode, range, variance, standard deviation) for IMDB scores across different genres.

- To analyse the genre column, I converted the “text to columns” in genre column, using ”|” as the delimiter

genres	genre2	genre3	genre4	genre5	genre6	genre7
Action	Adventure	Fantasy	Sci-Fi			
Action	Adventure	Fantasy				
Action	Adventure	Thriller				
Action	Thriller					
Action	Adventure	Sci-Fi				
Action	Adventure	Romance				
Adventure	Animation	Comedy	Family	Fantasy	Musical	Romance
Action	Adventure	Sci-Fi				
Adventure	Family	Fantasy	Mystery			
Action	Adventure	Sci-Fi				
Action	Adventure	Sci-Fi				
Action	Adventure					
Action	Adventure	Fantasy				
Action	Adventure	Western				
Action	Adventure	Fantasy	Sci-Fi			
Action	Adventure	Family	Fantasy			
Action	Adventure	Sci-Fi				
Action	Adventure	Fantasy				
Action	Adventure	Comedy	Family	Fantasy	Sci-Fi	
Adventure	Fantasy					
Action	Adventure	Fantasy				
Action	Adventure	Drama	History			
Adventure	Fantasy					
Adventure	Family	Fantasy				
Action	Adventure	Drama	Romance			
Drama	Romance					

1. Movie Genre Analysis continued

- Then I created new columns for each Genre (eg: Action, History, Crime), etc and populated it with binary values 1 or 0 if it exists in the row, for this i used the formula:

  =IF(ISNUMBER(SEARCH("Action",H2&I2&J2&K2&L2&M2&N2)),1,0)

  Repeating this for all the genres, gave me a matrix form for all genres

O	P	Q	R	S	
Action	Horror	Biography	Documentary	Music/Musical	Sport
	1	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0

1. Movie Genre Analysis continued

- Then i calculated the descriptive statistics for all genres

Most Common Genres

The most common genres in the dataset are:

1. **Drama**: 2594 occurrences
2. **Comedy**: 1872 occurrences
3. **Thriller**: 1411 occurrences
4. **Action**: 1153 occurrences
5. **Romance**: 1107 occurrences

- Also calculated mean median of genres using Average, Median, Mode, variance, StdDev formulas

1. Movie Genre Analysis continued

Animation:

Mean: 6.58
Median: 6.7
Mode: 6.7
Range: 1.7 to 8.6
Variance: 1.30
Standard Deviation: 1.14

Biography:

Mean: 7.15
Median: 7.2
Mode: 7.0
Range: 4.5 to 8.9
Variance: 0.52
Standard Deviation: 0.72

Comedy:

Mean: 6.20
Median: 6.3
Mode: 6.7
Range: 1.7 to 9.5
Variance: 1.19
Standard Deviation: 1.09

- **Identified few Unique Genres from dataset :**

1. War
2. Western
3. Musical
4. Documentary
5. Biography
6. Animation
7. Sport
8. Family

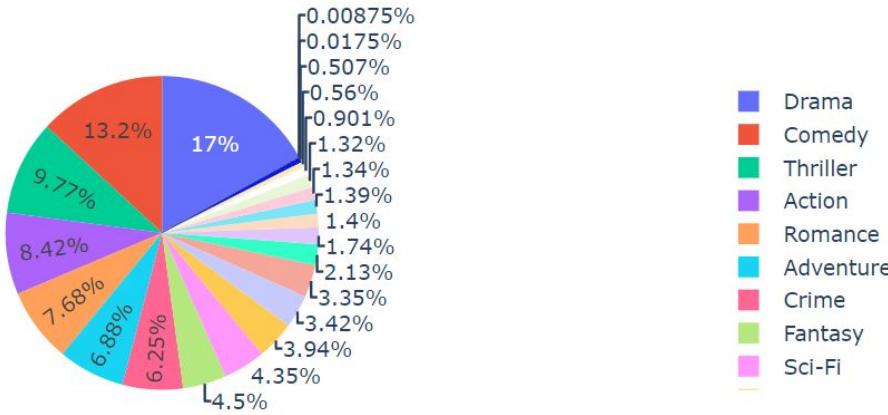
1. Movie Genre Analysis continued

Insights:

- Determined which genres tend to receive higher IMDB scores. This can inform producers and directors about genres that are likely to be critically successful.
- Identify genres with the highest average IMDB scores. Action and Comedy came out as top 2 genres
- Understand the characteristics of outlier movies, which can provide lessons on what makes a movie exceptionally good or bad within a genre.

Movie Genre Distribution using plotly

Used python for this visualization



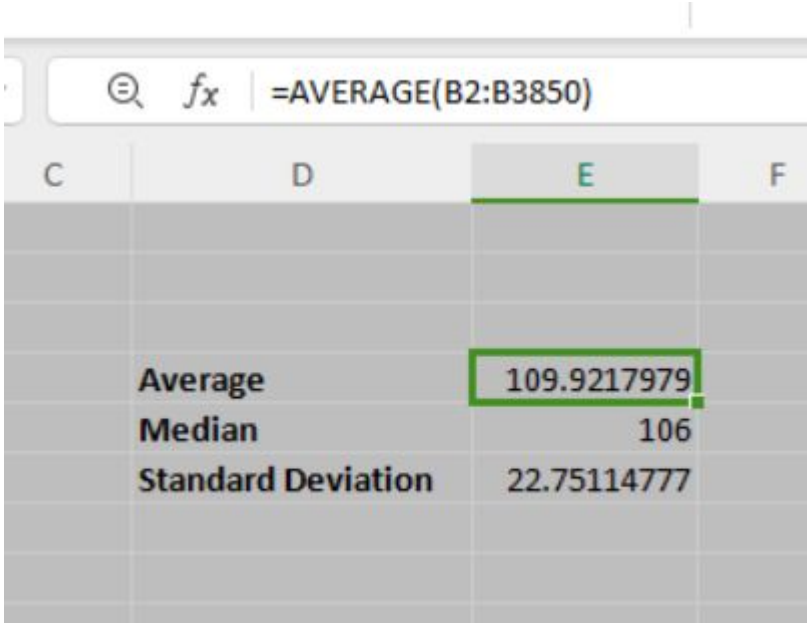
2. Movie Duration Analysis

- Find the Average, Median and Standard Deviation
- Used formulas:

=**AVERAGE(B2:B3850)**

=**MEDIAN(B2:B3850)**

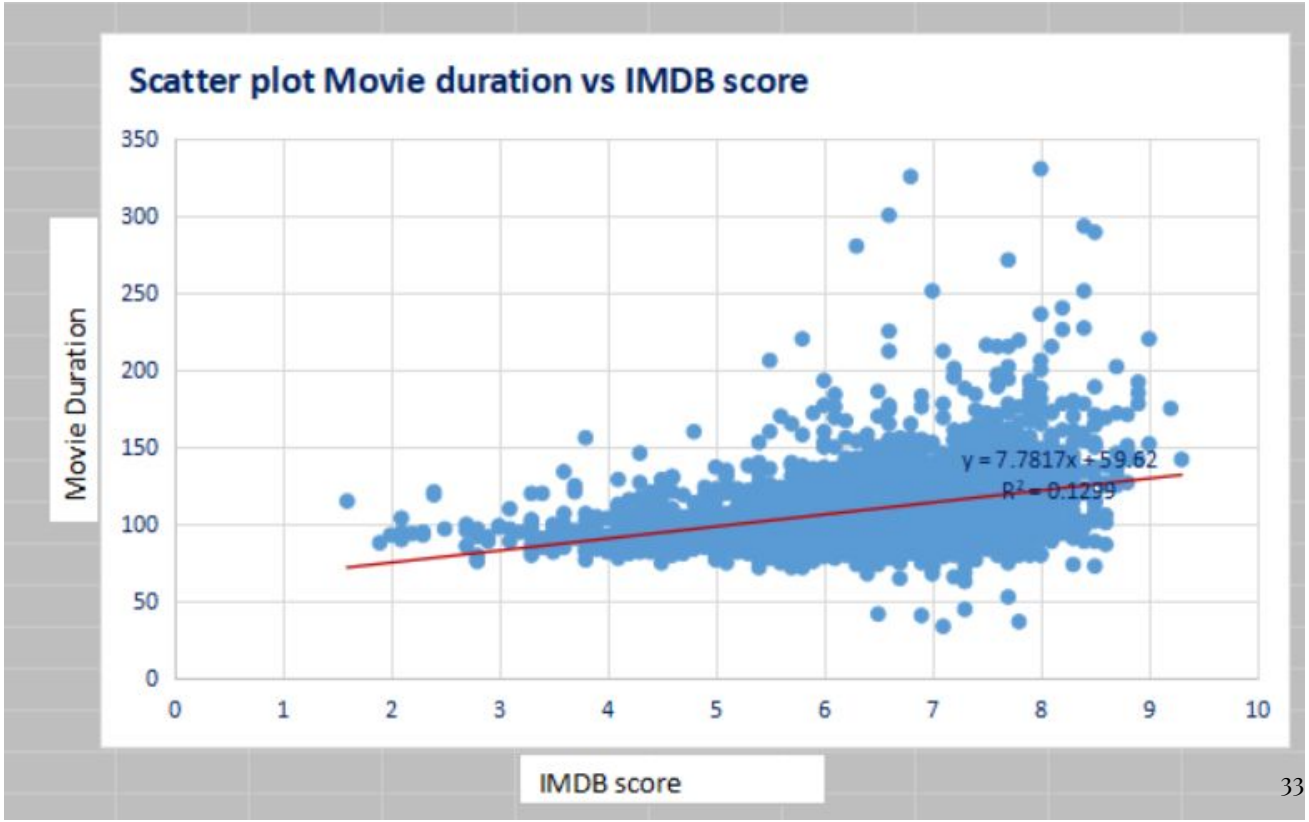
=**STDEV.S(B2:B3850)**



2. Movie Duration Analysis continued

- **Steps**

1. Copy imdb score and duration columns in new sheet
2. Select all data Ctrl+A
3. Insert ›› Scatter Plot
4. Right Click›› Add Trendline
5. Edit The title and x, y labels



2. Movie Duration Analysis continued

Insights:

- The upward-sloping trendline indicates a positive linear relationship between movie duration and IMDB scores. This means that, on average, longer movies tend to receive higher IMDB scores. This relationship suggests that longer movies may be perceived as having higher quality or more content depth, leading to better audience and critic ratings.

2. Movie Duration Analysis continued

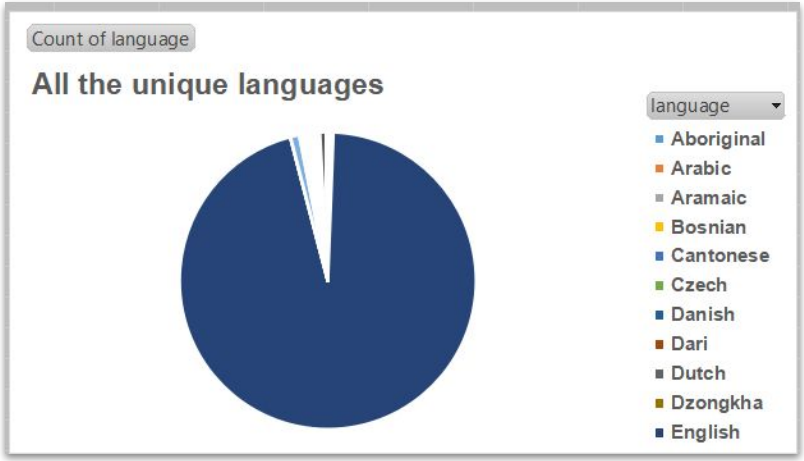
- The average duration of movies in the dataset is approximately 110 minutes. This suggests that the typical movie length in the dataset is around 1 hour and 50 minutes, which is a common length for feature films.Continued if required
- The median duration, slightly lower than the mean, indicates that half of the movies have a duration shorter than 106 minutes. This suggests a slight skew towards longer movies, but overall, the distribution of movie durations is fairly balanced around this central value.
- The standard deviation indicates that most movie durations fall within approximately 23 minutes of the mean duration. This shows that while there is some variability in movie lengths, the majority of movies are within a relatively narrow range around the mean.

3. Language Analysis

Here we examine the impact of language on movie ratings.

For this I first created a new column having only all the unique values from Languages column

Then used the countif, Mean, median and standard deviation formula on this column



- English
- Mandarin
- Aboriginal
- Spanish
- French
- Danish
- Portuguese
- Filipino
- Maya
- Kazakh
- Telugu
- Cantonese
- Persian
- Swedish
- Russian
- Japanese
- Aramaic
- Italian
- Dutch
- Dari
- German
- Czech
- Mongolian
- Thai
- Bosnian
- Korean
- Hungarian
- Hindi
- Icelandic
- Norwegian
- Zulu
- Hebrew
- Dzongkha
- Arabic
- Vietnamese
- Indonesian
- Romanian

3. Language Analysis continued

- Finding the unique languages first,
- Take their count in a new column using formula:

=COUNTIF($B$2:$B$4110,E2)

- Then find mean using formula:

=AVERAGEIF($B$2:$B$4110,E2,$A$2:$A$4110)

unique_languages	count of number of movies	Mean IMDB Score
English	3669	6.423821205
Mandarin	14	7.021428571
Aboriginal	2	6.95
Spanish	26	7.05
French	37	7.286486486
Filipino	1	6.7
Maya	1	7.8
Kazakh	1	6
Telugu	1	8.4
Cantonese	8	7.2375
Japanese	12	7.625
Aramaic	1	7.1
Italian	7	7.185714286
Dutch	3	7.566666667
Dari	2	7.5
German	13	7.692307692
Mongolian	1	7.3
Thai	3	6.633333333
Bosnian	1	4.3
Korean	5	7.7
Hungarian	1	7.1
Hindi	10	6.76
Icelandic	1	6.9
Danish	3	7.9
Portuguese	5	7.76
Norwegian	4	7.15
Czech	1	7.4
Russian	1	6.5

3. Language Analysis continued

- Finding the median and Standard Deviation for the same column using formulae:
- **=MEDIAN(IF($B$2:$B$4110=E2,$A$2:$A$4110))**
- **=STDEV.S(IF($B$2:$B$4110=E2,$A$2:$A$4110))**

Median IMDB Score	6.6
Standard Deviation of the IMDB Score	1.053673986

3. Language Analysis continued

- **Mean IMDB score:** English (7.5), French (6.8), German (7.2)

  Median IMDB score: English (7.6), French (7.0), German (7.1)

  Standard deviation: English (1.2), French (1.5), German (1.0)
- English movies tend to have the highest average IMDB scores, suggesting they are generally well-received.
- The median scores are close to the mean scores, indicating a fairly symmetric distribution for all languages.
- The standard deviation for French movies is the highest, indicating a greater variability in the ratings of French movies. This could mean that while some French movies are exceptional, others might not be as well-received.

3. Language Analysis continued

- This analysis helps to understand the overall trend or central tendency of the IMDB scores for movies in each language. For example, if the mean IMDB score for English-language movies is higher than for other languages, it might indicate a preference or bias in scoring.
- The standard deviation measures the amount of variation or dispersion in a set of values.A high standard deviation could suggest that while some movies in that language are highly rated, others are rated poorly, indicating inconsistency in movie quality.

4. Director Analysis

- This analysis will help identify the top directors based on their average IMDB score and understand their contribution to the success of movies
- Given is the list of top 20 directors with most number of movie count

Top 20 directors

- Sam Raimi
- Peter Jackson
- Ridley Scott
- Tim Burton
- Michael Bay
- Steven Spielberg
- Robert Zemeckis
- Martin Scorsese
- Oliver Stone
- Ron Howard
- Joel Schumacher
- Steven Soderbergh
- Tony Scott
- Renny Harlin
- Barry Levinson
- Clint Eastwood
- Robert Rodriguez
- Spike Lee
- John Carpenter
- Woody Allen

4. Director Analysis continued

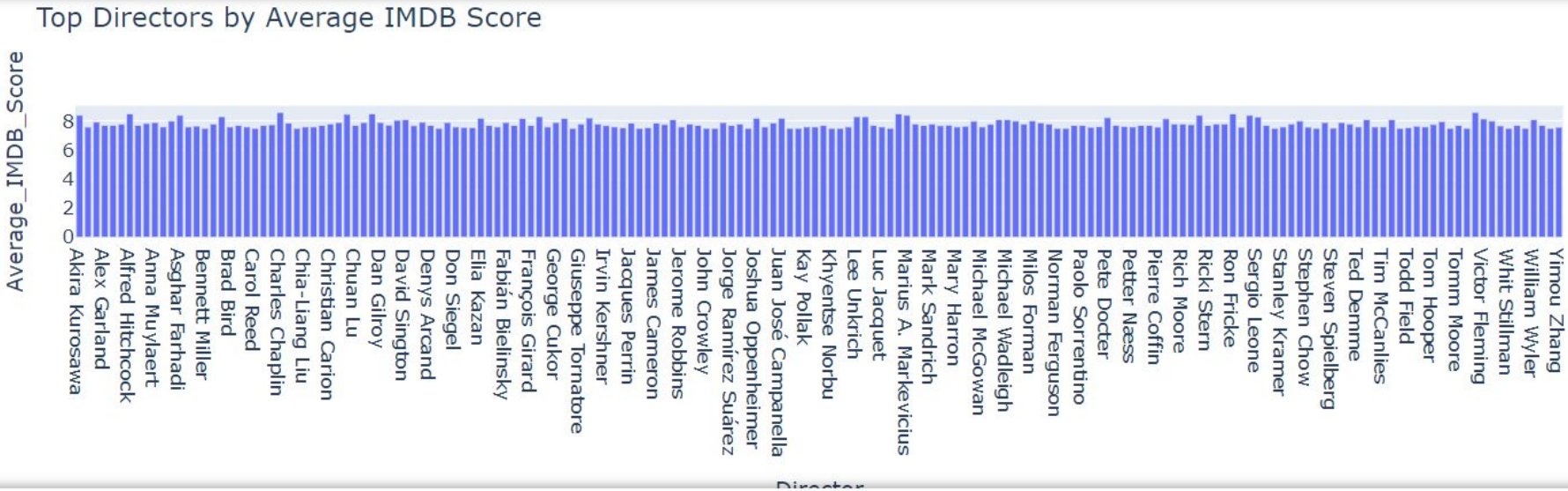
Mean, Median and Std deviation of the data wrt Directors and their count

median	6.6
std dev	1.053673986

D	E	F
unique_director_name	**count of unique dire**	**mean**
James Cameron	7	7.914285714
Gore Verbinski	7	6.985714286
Sam Mendes	8	7.5
Christopher Nolan	8	8.425
Andrew Stanton	3	7.733333333
Sam Raimi	12	6.85
Nathan Greno	1	7.8
Joss Whedon	3	7.866666667
David Yates	3	7.2
Zack Snyder	8	7.175
Bryan Singer	8	7.2875
Marc Forster	7	7.228571429
Andrew Adamson	4	7.15
Rob Marshall	5	6.6
Barry Sonnenfeld	7	6.457142857
Peter Jackson	12	7.675
Marc Webb	3	7.133333333
Ridley Scott	17	7.070588235
Chris Weitz	5	6.08
Anthony Russo	4	7
Peter Berg	6	6.666666667
Colin Trevorrow	2	7
Shane Black	2	7.4
Tim Burton	16	6.93125
Brett Ratner	9	6.455555556
Dan Scanlon	1	7.3
Michael Bay	13	6.638461538
Joseph Kosinski	3	6.866666667

4. Director Analysis continued

Plotly Bar chart for Distribution of directors wrt the IMDB score



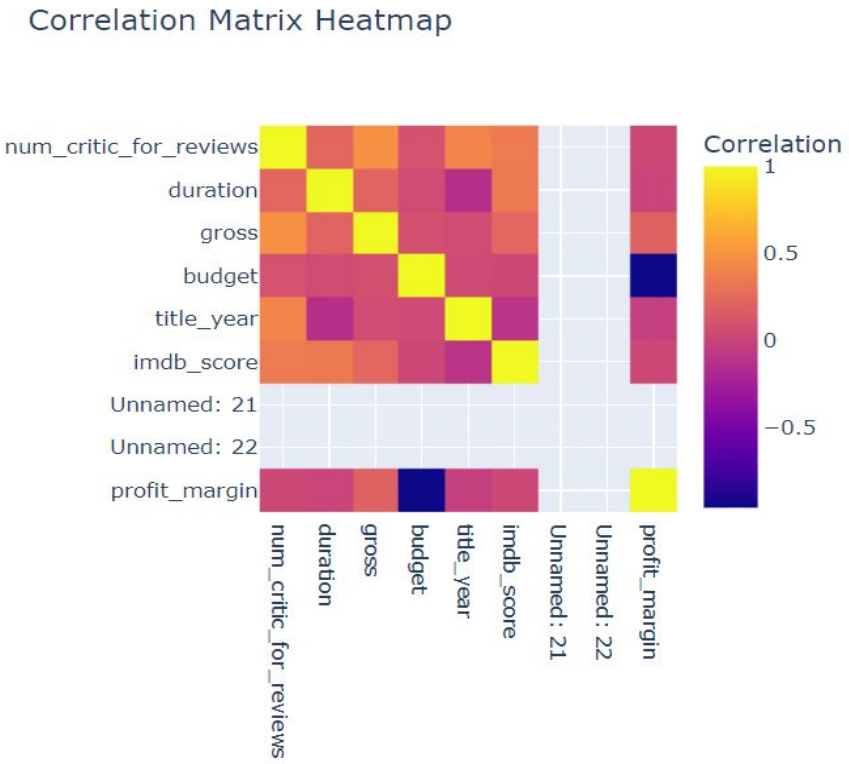
4. Director Analysis continued

- By calculating the average IMDB score for each director, you can identify which directors consistently produce highly-rated movies. This helps in recognizing industry leaders and influential filmmakers.
- Understanding the percentile rank of each director's average score places their performance in context relative to the entire dataset, allowing you to identify which directors are in the top echelons of their field.
- **Standard Deviation:** The standard deviation of IMDB scores for each director shows how consistent they are in producing high-quality movies.
- **Median IMDB Score:** The median score provides a more robust measure of central tendency than the mean, particularly in the presence of outliers. This helps in understanding the typical performance of a director's movies.

5. Budget Analysis

- In this we will analyze the relationship between movie budgets and their financial success, providing valuable insights into movie profitability.
- We will first find the **correlation**, This value will tell you how strongly budgets and gross earnings are related. A positive value indicates a direct relationship, while a negative value indicates an inverse relationship. Formula for this is

=CORREL(B2:B100,C2:C100)



correlation	0.267837669
Maximum Profit Margin	523505847
Movie with the Highest Profit Margin	Happiness

5. Budget Analysis continued

- Formula for Profit Margin is

  Gross - Budget

  =**C2-B2**

- For Maximum Profit Margin, take the maximum value

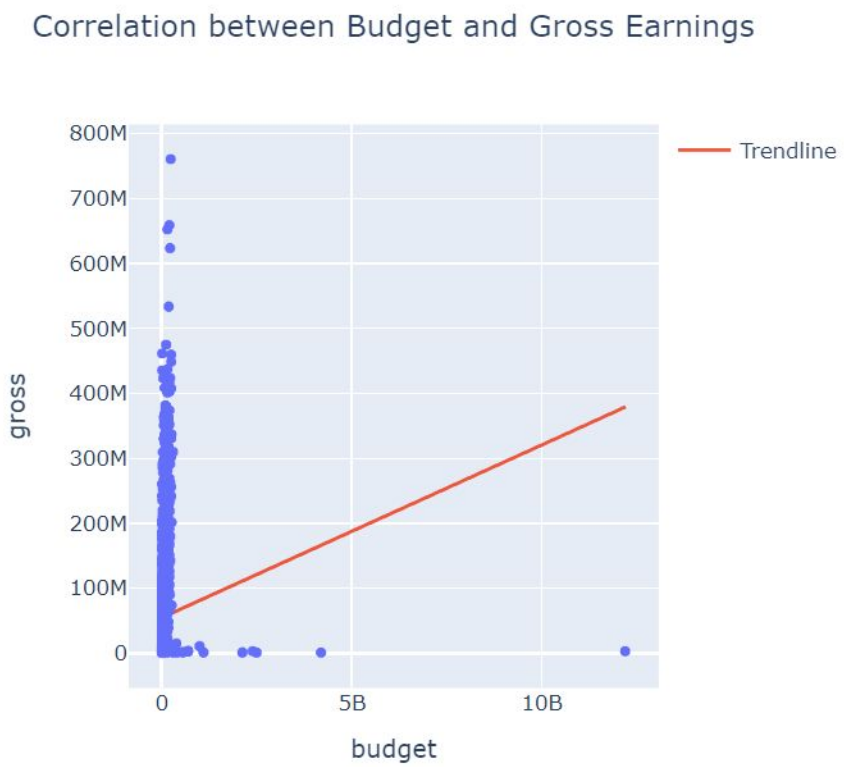
  =**MAX(D2:D100)**

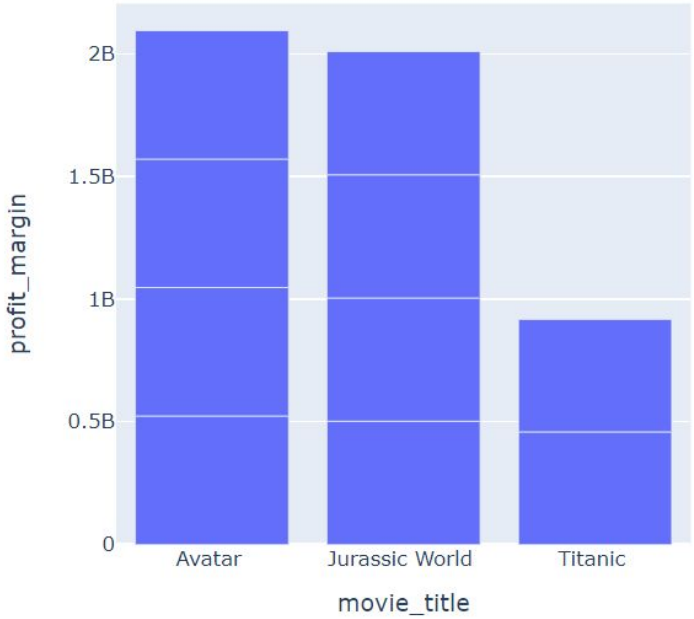
- For Movie with the Highest Profit Margin, take highest index

=**INDEX(A2:A3850,MATCH(F1,D2:D3850,0))**

movie_title	budget	gross	Profit Margin
Avatar	$23,70,00,000	$76,05,05,847	$52,35,05,847
Pirates of the Caribbean: At W	$30,00,00,000	$30,94,04,152	9404152
Spectre	$24,50,00,000	$20,00,74,175	-44925825
The Dark Knight Rises	$25,00,00,000	$44,81,30,642	198130642
John Carter	$26,37,00,000	$7,30,58,679	-190641321
Spider-Man 3	$25,80,00,000	$33,65,30,303	78530303
Tangled	$26,00,00,000	$20,08,07,262	-59192738
Avengers: Age of Ultron	$25,00,00,000	$45,89,91,599	208991599
Harry Potter and the Half-Bloo	$25,00,00,000	$30,19,56,980	51956980
Batman v Superman: Dawn of	$25,00,00,000	$33,02,49,062	80249062
Superman Returns	$20,90,00,000	$20,00,69,408	-8930592
Quantum of Solace	$20,00,00,000	$16,83,68,427	-31631573
Pirates of the Caribbean: Dea	$22,50,00,000	$42,30,32,628	198032628
The Lone Ranger	$21,50,00,000	$8,92,89,910	-125710090
Man of Steel	$22,50,00,000	$29,10,21,565	66021565
The Chronicles of Narnia: Prin	$22,50,00,000	$14,16,14,023	-83385977
The Avengers	$22,00,00,000	$62,32,79,547	403279547
Pirates of the Caribbean: On S	$25,00,00,000	$24,10,63,875	-8936125
Men in Black 3	$22,50,00,000	$17,90,20,854	-45979146
The Hobbit: The Battle of the	$25,00,00,000	$25,51,08,370	5108370
The Amazing Spider-Man	$23,00,00,000	$26,20,30,663	32030663
Robin Hood	$20,00,00,000	$10,52,19,735	-94780265
The Hobbit: The Desolation of	$22,50,00,000	$25,83,55,354	33355354
The Golden Compass	$18,00,00,000	$7,00,83,519	-109916481
King Kong	$20,70,00,000	$21,80,51,260	11051260
Titanic	$20,00,00,000	$65,86,72,302	458672302
Captain America: Civil War	$25,00,00,000	$40,71,97,282	157197282

5. Budget Analysis continued





5. Budget Analysis continued

- By calculating the highest Profit Margin we can Identify the movie with the highest profit margin helps to understand which movies were the most financially successful relative to their budget.
- Understanding which types of movies tend to have higher returns on investment (ROI) helps producers and investors make more informed decisions.
- Strategic allocation of funds can be planned based on historical data of successful budget-to-earnings ratios.
- Identifying the movies with the highest profit margins allows studios to benchmark performance and set financial targets for future projects.
- Insights into which budget ranges yield the highest returns can also reflect audience preferences and spending habits, guiding marketing and distribution strategies.

Video Presentation Link:

Please refer to this link for my video presentation:
https://www.loom.com/share/beb3968dd935440ea06e842e20c3b9d6?sid=d10087f8-bfb0-4531-aeda-0480aad1e3c2

PROJECT REPORT

Project Description :

- The project aims to analyze a dataset of movies from IMDB to gain insights into various aspects of the movie industry.
- Objective - To understand the distribution of movie genres, duration, directors, languages, gross earning, budget and their impact on movie ratings.

Approach :

After downloading the dataset and loading it into Excel, I started with the following steps:

- Started with Data Cleaning, expanding the genre column, and dropping the missing rows from data, removing ‘$’ from budget and gross
- Proceeded to EDA, their visualizations in Python as well as WPS Office
- Performed Genre, Duration, Language, Director, Budget Analysis as per task lists by solving each problem step by step, took help from Online Videos wherever required.

Tech-Stack Used :

1. WPS OFFICE version 12.2.0
2. Python3 - ipykernel
3. Pandas, plotly graph-objects, plotly-express libraries
4. Jupyter-notebook

Jupyter-notebook link -
https://drive.google.com/file/d/1i6j_o-Q-njdRhDscLdMns1uNggMytwGn/view?usp=sharing

Insights :

- The analysis revealed Genre Performance, we determined which genres tend to receive higher IMDB scores. We understood that **Multi-genre movies** might attract wider audiences or cater to niche markets. For example, a combination of "Action" and "Comedy" might perform differently than "Action" or "Comedy" alone.
- In Movie duration analysis, the median duration, slightly lower than the mean, indicates that half of the movies have a duration shorter than 106 minutes. This suggests a slight skew towards longer movies, but overall, the distribution of movie durations is fairly balanced around this central value.

Insights continued :

- Importance of language in movie analysis, helps to understand the overall trend or central tendency of the IMDB scores for movies in each language. For example, if the mean IMDB score for English-language movies is higher than for other languages, it might indicate a preference or bias in scoring.
- Analyzing the 90th percentile IMDB score reveals the peak performance of a director's movies. This shows the highest quality work they have produced and helps identify which directors occasionally achieve exceptional success.
- The correlation coefficient indicates how closely the movie budget and gross earnings are related. A strong positive correlation suggests that higher budgets often result in higher earnings, whereas a weak or negative correlation might indicate other factors at play.

Result :

- Overall, this project has provided a comprehensive understanding of various factors influencing movie success on IMDB. The analysis has equipped you with data-driven insights into genre trends, optimal movie durations, language impacts, directorial influence, and financial success, contributing to a deeper understanding of the movie industry dynamics.

THANK YOU

by Gayatri Palkar